

Article

Comparative Study on Energy Consumption of Neural Networks by Scaling of Weight-Memory Energy Versus Computing Energy for Implementing Low-Power Edge Intelligence

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Abstract

Energy consumption has emerged as a critical design constraint in deploying high-performance neural networks, especially on edge devices with limited power resources. In this paper, a comparative study is conducted for two prevalent deep learning paradigms—convolutional neural networks (CNNs), exemplified by ResNet18, and transformer-based large language models (LLMs), represented by GPT3-small, Llama-7B, and GPT3-175B. By analyzing how the scaling of memory energy versus computing energy affects the energy consumption of neural networks with different batch sizes (1, 4, 8, 16), it is shown that ResNet18 transitions from a memory energy-limited regime at low batch sizes to a computing energy-limited regime at higher batch sizes due to its extensive convolution operations. On the other hand, GPT-like models remain predominantly memory-bound, with large parameter tensors and frequent key–value (KV) cache lookups accounting for most of the total energy usage. Our results reveal that reducing weight-memory energy is particularly effective in transformer architectures, while improving multiply–accumulate (MAC) efficiency significantly benefits CNNs at higher workloads. We further highlight near-memory and in-memory computing approaches as promising strategies to lower data-transfer costs and enhance power efficiency in large-scale deployments. These findings offer actionable insights for architects and system designers aiming to optimize artificial intelligence (AI) performance under stringent energy budgets on battery-powered edge devices.

Keywords: energy consumption of neural networks; weight-memory energy; MAC computing energy; edge intelligence



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1. Introduction

As artificial intelligence (AI) technologies are advancing very rapidly nowadays, neural networks are starting to show that they can be really useful and helpful for various applications related to human cognition and intelligence [1–4]. The most representative models of neural networks can be regarded as convolutional neural networks and transformers [5–8]. They are used for processing spatial information such as images and sequential information such as natural languages, respectively, which can be considered the most common capabilities of human cognition and intelligence.

To realize human-level cognition and intelligence, very strong computing systems are needed. Figure 1 shows an entire hierarchy from Internet of Things (IoT) to cloud computing centers via an edge intelligence layer [9]. Here the IoT sensors collect massive